

Make VPC

- 4 subnets (1 public, 3 private)
- Enable in subnet settings public ip addresses
- Make it highly available (use 2 availability zones, the final private subnet can be the only one in a different subnet)
- Allocate an Elastic IP
- Create a nat gateway
- Create an internet gateway and attach it to your VPC
- Make route tables for your public and private subnets and attach an internet gateway and nat gateway to them respectively
- Make security groups for Bastion Host, web server, app server, and database
- Make sure to go back to security groups after making them and adding security groups to link them together, for example in the app server security group adding a rule for the database security group after creating the database security group.

Create Bastion Host

- EC2 instance
 - Amazon linux 2 ami
 - T2.micro
 - Use your vpc and public subnet
 - Use Security Group for Bastion Host made in VPC Setup

Create Web Server

- EC2 instance
 - Amazon linux 2 ami
 - T2.micro
 - Use your vpc and public subnet
 - In user data
 - `#!/bin/bash`
 - `sudo yum update -y`
 - `sudo amazon-linux-extras install -y lamp-mariadb10.2-php7.2 php7.2`
 - `Sudo yum install -y httpd`
 - `Use sudo systemctl start httpd to start up the webserver`
 - `Use sudo systemctl enable httpd to do it on reboot`
 - Use Security Group for Web Server made in VPC Setup

Create App Server

- EC2 instance
 - Amazon linux 2 ami
 - T2.micro
 - Use your vpc and public subnet
 - Type into user data
 - `#!/bin/bash`
 - `sudo yum install -y mariadb-server`
 - `Sudo service mariadb start`
 - Use Security Group for App Server made in VPC Setup

Create DB instance

- Create a subnet group
- Make a database instance
 - Standard create
 - mariadb
 - Free Tier
 - Disable automated backups
 - Disable encryption
 - User = root
 - Password = Re:Start!9
 - Initial database = mydb

Upload ssh keys to Bastion Host

- Windows users
 - Go to your command prompt and type out
 - `Pscp -scp -P 22 -i '.\Downloads\labsuser.ppk' -l user ec2-user '.\Downloads\labsuser.pem' ec2-user@bastion-host-public-ip:/home/ec2-user`
 - Replace labsuser.ppk and labsuser.pem with what you named your keys
 - Replace bastion-host-public-ip with your bastion host public ip address
- Mac and Linux Users
 - Go to your terminal and type out
 - `Chmod 400 labuser.pem`
 - `Scp -i '.\Downloads\labsuser.pem' -l user ec2-user '.\Downloads\labsuser.pem' ec2-user@bastion-host-public-ip:/home/ec2-user`
 - Replace labsuser.ppk and labsuser.pem with what you named your keys
 - Replace bastion-host-public-ip with your bastion host public ip address

Test to make sure key was uploaded into Bastion Host

- Ssh into Bastion Host
- Type ls
- Labsuser.pem should show up

Connect to App Server

- From ssh inside of Bastion Host use the terminal
- Type out `chmod 400 labsuser.pem` to change file permissions
- Type out `ssh -i my-key-pair.pem ec2-user@app-server-private-ip`
- Replace my-key-pair with the name of your key
- Replace app-server-private-ip with your app server's private ip address
- Test out pinging the web server by typing out `ping` and the private ip address
- Test out connecting to database by typing out `mysql -user =root -password='Re:Start!9' -host=database-server-endpoint`

- Replace database-server-endpoint with the database server endpoint
- Type show databases; to see your database from the app server